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A

Using $TE = -\frac{GMm}{2r}$, $KE = \frac{GMm}{2r}$, we see that $KE = -TE$. Satellites with same KE has the same TE. Option A is correct.

Using $KE = \frac{GMm}{2r}$, $GPE = -\frac{GMm}{r}$, we see that $KE = -\frac{1}{2} GPE$. Satellites with the same KE has the same GPE. Options B and C are wrong.

Using $KE = \frac{GMm}{2r}$, we see that satellite A with larger mass must have a larger orbital radius than satellite B. Using T^2 proportional to r^3 and $\omega = 2\pi/T$, satellite A with larger r will have larger T , hence smaller ω . Option D is wrong.

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A